GEOMETRY · TRANSFORMATIONS

Transformation Studio

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Transformation Studio** on Mr. K's Classroom: Geometry → Transformations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. What do you think happens to (2, 5) when it is reflected over the x-axis?

   It becomes (2, –5).

2 Explore use the tool and record what you see

Transform △ABC with A(1, 1), B(5, 2), C(2, 5)

Transformation	Rule	A′	B′	C′
Translate 3 right, 2 down	**(x + 3, y – 2)**	**(4, –1)**	**(8, 0)**	**(5, 3)**
Reflect over the x-axis	**(x, –y)**	**(1, –1)**	**(5, –2)**	**(2, –5)**
Reflect over y = x	**(y, x)**	**(1, 1)**	**(2, 5)**	**(5, 2)**
Rotate 90° counterclockwise	**(–y, x)**	**(–1, 1)**	**(–2, 5)**	**(–5, 2)**
Rotate 180°	**(–x, –y)**	**(–1, –1)**	**(–5, –2)**	**(–2, –5)**
Dilate by 2	**(2x, 2y)**	**(2, 2)**	**(10, 4)**	**(4, 10)**

What each transformation keeps

Transformation	Image congruent?	Orientation kept?
Translation	**yes**	**yes**
Reflection	**yes**	**no**
Rotation	**yes**	**yes**
Dilation (k ≠ 1)	**no (similar)**	**yes**

3 Write the rule fill in each blank from what you observed

1. Translations, reflections and rotations are **rigid** motions: the image is **congruent** to the preimage.
2. A reflection **reverses** the orientation. A rotation **keeps** it.
3. Rotating 90° counterclockwise about the origin: (x, y) → **(–y, x)**.
4. Reflecting over y = x: (x, y) → **(y, x)**.
5. A dilation with k ≠ 1 makes an image that is **similar**, not congruent.

4 Check no tool: show your work

1. Rotate (–3, 4) 180° about the origin.

   (3, –4)

2. A translation moves (2, 1) to (–1, 5). Write its rule.

   (x – 3, y + 4)

3. Reflect (6, –2) over the y-axis.

   (–6, –2)

4. Rotate (4, 1) 270° counterclockwise about the origin.

   (1, –4)

5 Reflect

How can you tell from the vertex order that a figure was reflected?

A → B → C runs clockwise in one figure and counterclockwise in the other.